AKSHAY MOHAN M P

BACKEND ENGINEER (GO) | IoT PLATFORMS | AWS & DEVOPS

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PROFESSIONAL SUMMARY

Backend engineer with 2+ years of production experience in **Go**, building multi-tenant IoT and laboratory platforms, device-security services and the **AWS / Terraform / CI-CD** infrastructure behind them. Led a full Express.js-to-Go migration; also work across edge software (Raspberry Pi), Python and TypeScript for computer-vision pipelines. Biomedical-engineering background brings a hardware-aware, safety-conscious approach.

TECHNICAL SKILLS

Languages	Go (primary), Python, TypeScript / JavaScript, SQL, Bash
Backend & APIs	Gin, REST / OpenAPI, JWT, RBAC, OpenFGA, pgx, Protobuf, Node.js / Express.js, microservices
Data & Messaging	PostgreSQL, PGMQ, SQLite, MQTT (VerneMQ, EMQX), Firebase Cloud Messaging, AWS SQS, S3 / MinIO, ClickHouse
Cloud & DevOps	AWS (EC2, Lightsail, S3, SES, Secrets Manager, IAM Identity Center, Amplify, CloudFront), Terraform / OpenTofu, Docker, GitHub Actions, Coolify, Nomad, Nginx, Linux
Security	X.509 PKI, mutual TLS, TLS automation, secrets management, private registry (Zot), VPN (NetBird)
Observability & Testing	OpenTelemetry, HyperDX, structured logging, Go testing (testify, pgxmock), Artillery load testing
Frontend & ML Tooling	React, Redux Toolkit, MUI; CVAT, ClearML, PyTorch, OpenCV

PROFESSIONAL EXPERIENCE

Software Engineer (Backend, Go) | Sahas Labs, Kolhapur, Maharashtra — Aug 2024 – Present

- **Backend & APIs:** Led the migration of the core multi-tenant IoT backend from Express.js to Go and own two Go platforms (device-fleet management and laboratory/test management; ~180 documented API paths, 600+ automated tests) built on Gin, PostgreSQL/pgx and OpenAPI docs.
- **Auth & Access Control:** Designed tenant-scoped JWT authentication, four-tier RBAC (40+ permissions) and OpenFGA relationship checks so every request resolves to the right company, plant or machine.
- **Ingestion & Messaging:** Built bulk, de-duplicated, rate-limited device-measurement ingestion and a PGMQ-backed alert worker that sends push notifications through Firebase Cloud Messaging (migrated from AWS SQS).
- **Device Security & PKI:** Built a Go certificate service (internal CA, per-device X.509, CRL, bulk renewal) with MQTT auth hooks for VerneMQ and EMQX; run a self-hosted Zot container registry secured with mutual TLS.
- **Edge & Embedded IoT:** Co-developed Go software for a Raspberry Pi 5 test bench (Modbus TCP PLC control, OpenCV measurements, SQLite, live dashboard) and built its signed over-the-air update system and MQTT command channel.
- **ERP & Reporting:** Integrated Odoo ERP with the lab platform through dedicated API routes and generated calibration reports as PDFs.
- **Computer Vision & MLOps:** Extended a self-hosted CVAT fork (Python, TypeScript/React) with temporal labelling, auto-annotation and ClearML dataset lineage; built an unattended AWS spot-GPU training loop (~$0.40 per detector run) and trained D-FINE / Mask R-CNN models on plant-camera footage.
- **Cloud Tooling:** Wrote a single-binary Go dashboard that drives OpenTofu to provision EC2, Lightsail, RDS, S3 and ALB/CloudFront, with IAM Identity Center SSO, cost breakdowns, CloudWatch metrics and SSM commands.
- **DevOps & Observability:** Run production on AWS with Terraform/OpenTofu, Docker and GitHub Actions (test, ARM64 build, deploy); self-host Coolify, Gitea, GitHub runners and a NetBird VPN; instrument services with OpenTelemetry; load-test with Artillery and deliver capacity reports.
- **Also:** Ported a Python (FastAPI) backend to Go, built the Hairlytics API (Gin, JWT, pgx, MinIO) and established Go testing and code-review practices.

Web Developer Intern | Dev Town — Mar 2024 – Jun 2024

- Built Express.js REST APIs and backend microservices, deployed staging environments on AWS Amplify and documented endpoints for internal developers.

Biomedical Engineer | National Health Mission & Baby Memorial Hospital — May 2019 – Oct 2021

- Maintained and safety-tested diagnostic, patient-monitoring and surgical equipment across state health networks; moved into software while completing a BTech.

EDUCATION

BTech, Electrical, Electronics & Communications Engineering — 2020 – 2024
APJ Abdul Kalam Technological University, Kerala

Diploma in Engineering, Biomedical Engineering — 2016 – 2019
Board of Technical Education, Kerala

Languages: English (Professional) | Hindi (Professional) | Malayalam (Native)